

UNIT 1:

Introduction to Database Systems:

Introduction:

In computerized information system data are the basic resource of the organization. So, proper organization and management for data is required for organization to run smoothly. Database management system deals the knowledge of how data stored and managed on a computerized information system. In any organization, it requires accurate and reliable data for better decision making, ensuring privacy of data and controlling data efficiently.

The examples include deposit and/or withdrawal from a bank, hotel, airline or railway reservation,

purchase items from supermarkets in all cases, a database is accessed.

What is data?

Data are the known facts or figures that have implicit meaning. It can also be defined as it is the representation of facts, concepts or instructions in a formal manner, which is suitable for understanding and processing. Data can be represented in alphabets (A-Z, a-z), digits (0-9) and using special characters (+, -, #, \$, etc)

e.g: 25, "ajit" etc.

Information:

Information is the processed data on which decisions and actions are based. Information can be defined as the organized and classified data to provide meaningful values.

Eg: "The age of Ravi is 25"

File:

File is a collection of related data stored in secondary memory.

File Oriented Approach:

The traditional file oriented approach to information processing each application has a separate master file and its own set of personal file. In file oriented approach the program dependent on the files and files dependent upon the programs.

Disadvantages of file oriented approach:

1) Data redundancy and inconsistency:

The same information may be written in several files. This redundancy leads to higher storage and access cost. It may lead data inconsistency that is the various copies of the same data may present at multiple places for example a changed customer address may be reflected in single file but not else where in the system.

2) Difficulty in accessing data :

The conventional file processing system do not allow data to be retrieved in a convenient and efficient manner according to user choice.

3) Data isolation :

Because data are scattered in various files and files may be in different formats with new application programs to retrieve the appropriate data is difficult.

4) Integrity Problems:

Developers enforce data validation in the system by adding appropriate code in the various application program. However when new constraints are added, it is difficult to change the programs to enforce them.

5) Atomicity:

It is difficult to ensure atomicity in a file processing system when transaction failure occurs due to power failure, networking problems etc. (atomicity: either all operations of the transaction are reflected properly in the database or non are)

6) Concurrent access: In the file processing system it is not possible to access the same file for transaction at same the time.

7) Security problems: There is no security provided in file processing system to secure the data from unauthorized user access.

Database:

A database is organized collection of related data of an organization stored in formatted way which is shared by multiple users.

The main feature of data in a database are:

1. It must be well organized
2. It is related
3. It is accessible in a logical order without any difficulty
4. It is stored only once

For example consider the roll no, name, address of a student stored in a student file. It is collection of related data with an implicit meaning. Data in the database may be persistent, integrated and shared.

Persistent:

If data is removed from database due to some explicit request from user to remove.

Integrated:

A database can be a collection of data from different files and when any redundancy among those files are removed from database is said to be integrated data.

Sharing Data:

The data stored in the database can be shared by multiple users simultaneously without affecting the correctness of data.

Why Database:

In order to overcome the limitation of a file system, a new approach was required. Hence a database approach emerged. A database is a persistent collection of logically related data. The initial attempts were to provide a centralized collection of data. A database has a self-describing nature. It contains not only the data sharing and integration of data of an organization in a single DBMS

database. A small database can be handled manually but for a large database and having multiple users it is difficult to maintain it. In that case a computerized database is useful.

The advantages of database system over traditional, paper-based methods of record keeping are:

- **Compactness:** No need for large amount of paper files
- **Speed:** The machine can retrieve and modify the data faster way than human being
- **Less drudgery:** Much of the maintenance of files by hand is eliminated
- **Accuracy:** Accurate, up-to-date information is fetched as per requirement of the user at any time.

Database Management System (DBMS):

A database management system consists of collection of related data and refers to a set of programs for defining, creation, maintenance and manipulation of a database.

Function of DBMS:

1. Defining database schema: it must give facility for defining the database structure also specifies access rights to authorized users.
2. Manipulation of the database: The dbms must have functions like insertion of record into database, updation of data, deletion of data, retrieval of data.
3. Sharing of database: The DBMS must share data items for multiple users by maintaining consistency of data.
4. Protection of database: It must protect the database against unauthorized users.
5. Database recovery: If for any reason the system fails DBMS must facilitate data base recovery.

Advantages of DBMS:

Reduction of redundancies:

Centralized control of data by the DBA avoids unnecessary duplication of data and effectively reduces the total amount of data storage required avoiding duplication in the elimination of the inconsistencies that tend to be present in redundant data files.

Sharing of Data:

A database allows the sharing of data under its control by any number of application programs or users.

Data Integrity:

Data integrity means that the data contained in the database is both accurate and consistent. Therefore data values being entered for storage could be checked to ensure that they fall within a specified range and are of the correct format.

Data Security:

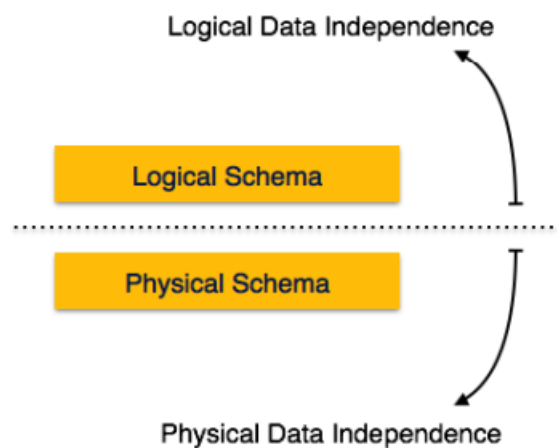
The DBA who has the ultimate responsibility for the data in the dbms can ensure that proper access procedures are followed including proper authentication to access to the DataBase System and additional check before permitting access to sensitive data.

Conflict Resolution:

DBA resolve the conflict on requirements of various user and applications. The DBA chooses the best file structure and access method to get optional performance for the application.

Data Independence:

- A database system normally contains a lot of data in addition to users' data. For example, it stores data about data, known as metadata, to locate and retrieve data easily.
- It is rather difficult to modify or update a set of metadata once it is stored in the database.
- But as a DBMS expands, it needs to change over time to satisfy the requirements of the users. If the entire data is dependent, it would become a tedious and highly complex job.



- Metadata itself follows a layered architecture, so that when we change data at one layer, it does not affect the data at another level. This data is independent but mapped to each other.

Data independence is usually considered from two points of views; physically data independence and logical data independence.

Physical Data Independence allows changes in the physical storage devices or organization of the files to be made without requiring changes in the conceptual view or any of the external views and hence in the application programs using the data base.

- All the schemas are logical, and the actual data is stored in bit format on the disk. Physical data independence is the power to change the physical data without impacting the schema or logical data.
- For example, in case we want to change or upgrade the storage system itself – suppose we want to replace hard-disks with SSD – it should not have any impact on the logical data or schemas.

Logical Data Independence indicates that the conceptual schema can be changed without affecting the existing external schema or any application program.

- Logical data is data about database, that is, it stores information about how data is managed inside. For example, a table (relation) stored in the database and all its constraints, applied on that relation.
- Logical data independence is a kind of mechanism, which liberalizes itself from actual data stored on the disk. If we do some changes on table format, it should not change the data residing on the disk.

Disadvantage of DBMS:

1. DBMS software and hardware (networking installation) cost is high
2. The processing overhead by the dbms for implementation of security, integrity and sharing of the data.
3. Centralized database control
4. Setup of the database system requires more knowledge, money, skills, and time.
5. The complexity of the database may result in poor performance.

DBMS	File System
DBMS is a collection of data. In DBMS, the user is not required to write the procedures.	File system is a collection of data. In this system, the user has to write the procedures for managing the database.
Searching data is easy in Dbms	Searching is difficult in File System
Dbms is structured data	Files are unstructured data
No data redundancy in Dbms	Data redundancy is there in file system
Memory utilisation well in dbms	Memory utilisation poor in file system
No data inconsistency in dbms	Inconsistency in file system

DBMS gives an abstract view of data that hides the details.	File system provides the detail of the data representation and storage of data.
DBMS provides a crash recovery mechanism, i.e., DBMS protects the user from the system failure.	File system doesn't have a crash mechanism, i.e., if the system crashes while entering some data, then the content of the file will lost.
DBMS provides a good protection mechanism.	It is very difficult to protect a file under the file system.
DBMS contains a wide variety of sophisticated techniques to store and retrieve the data.	File system can't efficiently store and retrieve the data.
DBMS takes care of Concurrent access of data using some form of locking.	In the File system, concurrent access has many problems like redirecting the file while other deleting some information or updating some information.

- A DBMS is software that allows creation, definition and manipulation of database, allowing users to store, process and analyse data easily.

- DBMS provides us with an interface or a tool, to perform various operations like creating database, storing data in it, updating data, creating tables in the database and a lot more.
- DBMS also provides protection and security to the databases.
- It also maintains data consistency in case of multiple users.

Here are some examples of popular DBMS used these days:

- MySql
- Oracle
- SQL Server
- IBM DB2

DATABASE APPLICATIONS – DBMS:

- Applications where we use Database Management Systems are:
- Telecom: There is a database to keeps track of the information regarding calls made, network usage, customer details etc.
- Industry: Where it is a manufacturing unit, warehouse or distribution centre, each one needs a database to keep the records of ins and outs
- Banking System: For storing customer info, tracking day to day credit and debit transactions, generating bank statements etc.
- Sales: To store customer information, production information and invoice details.
- Airlines: To travel though airlines, we make early reservations; this reservation information along with flight schedule is stored in database.
- Education sector: Database systems are frequently used in schools and colleges to store and retrieve the data regarding student details, staff details, course details, exam details, payroll data, attendance details, fees details etc.

PURPOSE OF DATABASE SYSTEMS

The main purpose of database systems is to manage the data. Consider a university that keeps the data of students, teachers, courses, books etc. To manage this data we need to store this data somewhere where we can add new data, delete unused data, update outdated data, retrieve data, to perform these operations on data we need a Database management system that allows us to store the data in such a way so that all these operations can be performed on the data efficiently.

Characteristics of DBMS

- Data stored into Tables: Data is never directly stored into the database. Data is stored into tables, created inside the database.
- Reduced Redundancy: In the modern world hard drives are very cheap, but earlier when hard drives were too expensive, unnecessary repetition of data in database was a big problem. But DBMS follows Normalisation which divides the data in such a way that repetition is minimum.
- Data Consistency: On Live data, i.e. data that is being continuously updated and added, maintaining the consistency of data can become a challenge. But DBMS handles it all by itself.

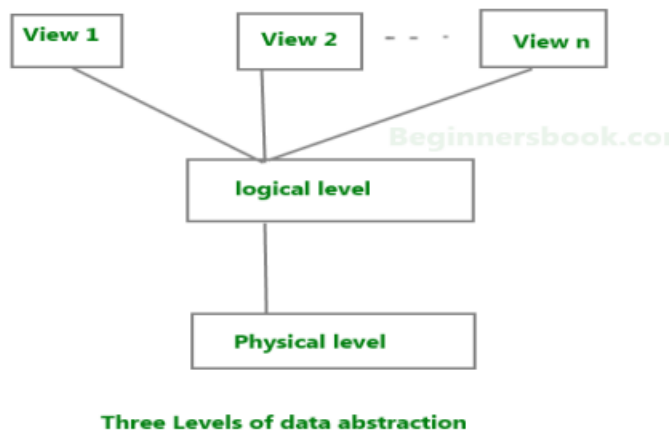
- **Support Multiple user and Concurrent Access:** DBMS allows multiple users to work on it(update, insert, delete data) at the same time and still manages to maintain the data consistency.
- **Query Language:** DBMS provides users with a simple Query language, using which data can be easily fetched, inserted, deleted and updated in a database.

View of Data in DBMS

- Abstraction is one of the main features of database systems.
- Hiding irrelevant details from user and providing abstract view of data to users, helps in easy and efficient user-database interaction.
- the three level of DBMS architecture, The top level of that architecture is “view level”. The view level provides the “view of data” to the users and hides the irrelevant details such as data relationship, database schema, constraints, security etc from the user.

Data Abstraction in DBMS

Database systems are made-up of complex data structures. To ease the user interaction with database, the developers hide internal irrelevant details from users. This process of hiding irrelevant details from user is called data abstraction.



We have three levels of abstraction:

Physical level: This is the lowest level of data abstraction. It describes how data is actually

stored in database. You can get the complex data structure details at this level.

Logical level: This is the middle level of 3-level data abstraction architecture. It describes

what data is stored in database.

View level: Highest level of data abstraction. This level describes the user interaction with database system.

Instance and schema in DBMS

- **Definition of schema:** Design of a database is called the schema. Schema is of three types: Physical schema, logical schema and view schema.
- The design of a database at physical level is called physical schema, how the data stored in blocks of storage is described at this level.
- Design of database at logical level is called logical schema, programmers and database administrators work at this level, at this level data can be described as certain types of data records gets stored in data structures, however the internal details such as implementation of data structure is hidden at this level (available at physical level).

- Design of database at view level is called view schema. This generally describes end user interaction with database systems.

Definition of instance:

The data stored in database at a particular moment of time is called instance of database. Database schema defines the variable declarations in tables that belong to a particular database; the value of these variables at a moment of time is called the instance of that database.

DBMS ARCHITECTURE:

- Database management systems architecture will help us understand the components of database system and the relation among them.
- The architecture of DBMS depends on the computer system on which it runs.
- the basic client/server architecture is used to deal with a large number of PCs, web servers, database servers and other components that are connected with networks.
- The client/server architecture consists of many PCs and a workstation which are connected via the network.
- DBMS architecture depends upon how users are connected to the database to get their request done.

TYPES OF DBMS ARCHITECTURE

There are three types of DBMS architecture:

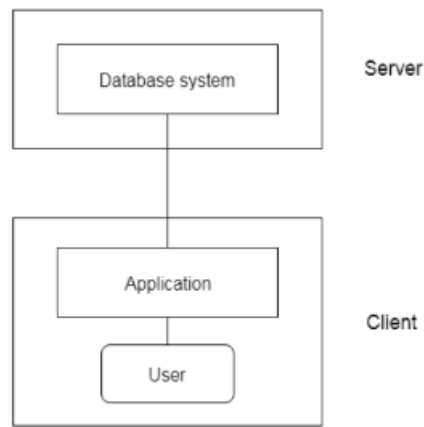
1. Single tier architecture
2. Two tier architecture
3. Three tier architecture

1-Tier Architecture

- In this type of architecture, the database is readily available on the client machine, any request made by client doesn't require a network connection to perform the action on the database.
- Any changes done here will directly be done on the database itself. It doesn't provide a handy tool for end users.
- The 1-Tier architecture is used for development of the local application, where programmers can directly communicate with the database for the quick response.

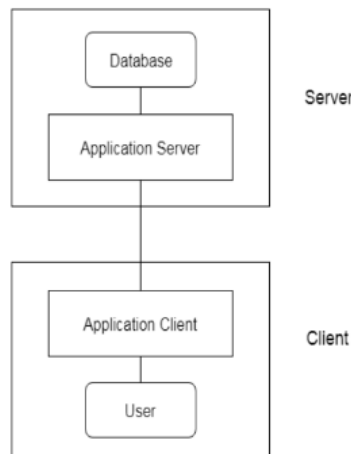
2. Two tier architecture

- In two-tier architecture, the Database system is present at the server machine and the DBMS application is present at the client machine, these two machines are connected with each other through a reliable network.
- Whenever client machine makes a request to access the database present at server using a query language like sql, the server perform the request on the database and returns the result back to the client.
- The application connection interface such as JDBC, ODBC are used for the interaction between server and client.



3. 3-Tier Architecture

- In three-tier architecture, another layer is present between the client machine and server machine.
- In this architecture, the client application doesn't communicate directly with the database systems present at the server machine, rather the client application communicates with server application and the server application internally communicates with the database system present at the server.

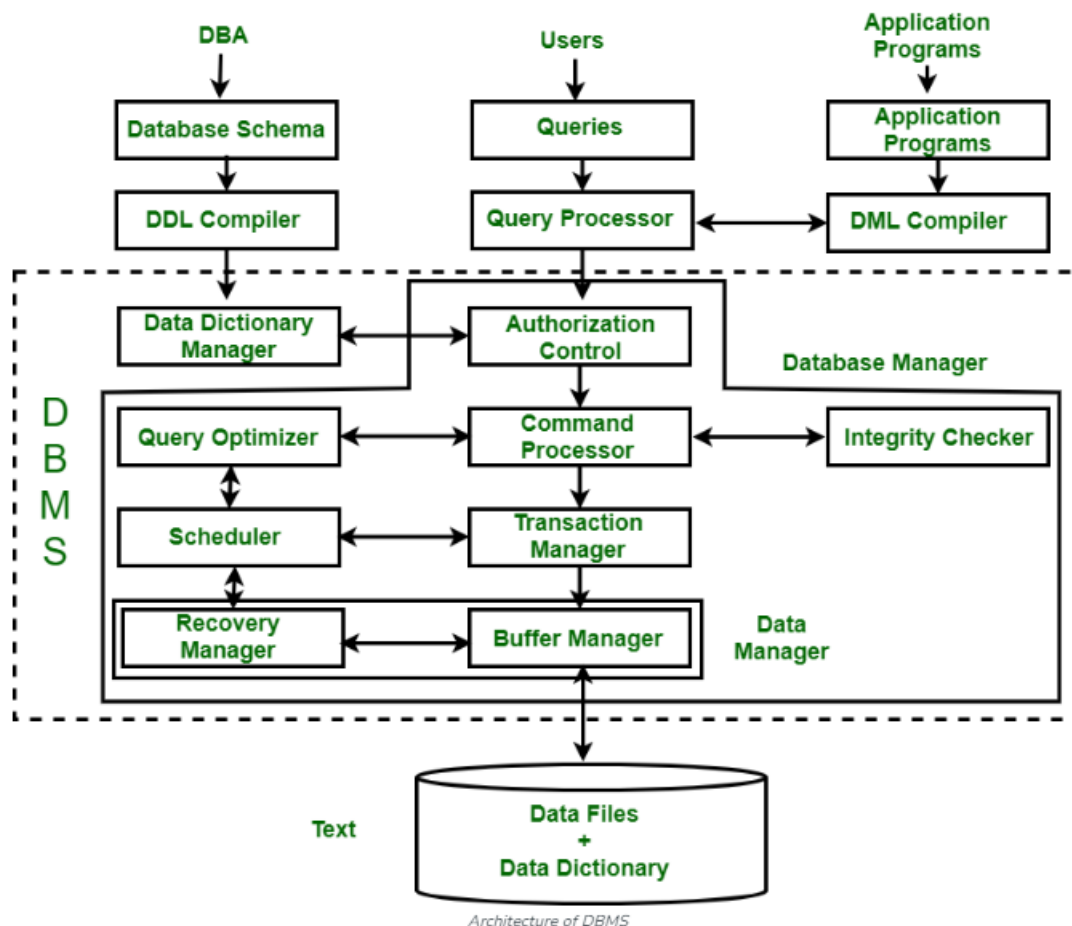


Structure of Database Management System

Database Management System (DBMS) is software that allows access to data stored in a database and provides an easy and effective method of –

- Defining the information.
- Storing the information.
- Manipulating the information.
- Protecting the information from system crashes or data theft.
- Differentiating access permissions for different users.

The database system is divided into three components: Query Processor, Storage Manager, and Disk Storage. These are explained as following below.



1. Query Processor: It interprets the requests (queries) received from end user via an application program into instructions. It also executes the user request which is received from the DML compiler.

Query Processor contains the following components –

- **DML Compiler:** It processes the DML statements into low level instruction (machine language), so that they can be executed.
- **DDL Interpreter:** It processes the DDL statements into a set of table containing meta data (data about data).
- **Embedded DML Pre-compiler:** It processes DML statements embedded in an application program into procedural calls.
- **Query Optimizer:** It executes the instruction generated by DML Compiler.

2. Storage Manager: Storage Manager is a program that provides an interface between the data stored in the database and the queries received. It is also known as Database Control System. It maintains the consistency and integrity of the database by applying the constraints and executing the DCL statements. It is responsible for updating, storing, deleting, and retrieving data in the database.

It contains the following components –

- **Authorization Manager:** It ensures role-based access control, i.e., checks whether the particular person is privileged to perform the requested operation or not.
- **Integrity Manager:** It checks the integrity constraints when the database is modified.
- **Transaction Manager:** It controls concurrent access by performing the operations in a scheduled way that it receives the transaction. Thus, it ensures

that the database remains in the consistent state before and after the execution of a transaction.

- **File Manager:** It manages the file space and the data structure used to represent information in the database.
- **Buffer Manager:** It is responsible for cache memory and the transfer of data between the secondary storage and main memory.

3. Disk Storage: It contains the following components –

- **Data Files:** It stores the data.
- **Data Dictionary:** It contains the information about the structure of any database object. It is the repository of information that governs the metadata.
- **Indices:** It provides faster retrieval of data item.

The structure of a Database Management System (DBMS) can be divided into three main components: the Internal Level, the Conceptual Level, and the External Level.

1. **Internal Level:** This level represents the physical storage of data in the database. It is responsible for storing and retrieving data from the storage devices, such as hard drives or solid-state drives. It deals with low-level implementation details such as data compression, indexing, and storage allocation.
2. **Conceptual Level:** This level represents the logical view of the database. It deals with the overall organization of data in the database and the relationships between them. It defines the data schema, which includes tables, attributes, and their relationships. The conceptual level is independent of any specific DBMS and can be implemented using different DBMSs.
3. **External Level:** This level represents the user's view of the database. It deals with how users access the data in the database. It allows users to view data in a way that makes sense to them, without worrying about the underlying implementation details. The external level provides a set of views or interfaces to the database, which are tailored to meet the needs of specific user groups.

The three levels are connected through a schema mapping process that translates data from one level to another. The schema mapping process ensures that changes made at one level are reflected in the other levels.

In addition to these three levels, a DBMS also includes a Database Administrator (DBA) component, which is responsible for managing the database system. The DBA is responsible for tasks such as database design, security management, backup and recovery, and performance tuning.

Overall, the structure of a DBMS is designed to provide a high level of abstraction to users, while still allowing low-level implementation details to be managed effectively. This allows users to focus on the logical organization of data in the database, without worrying about the physical storage or implementation details.